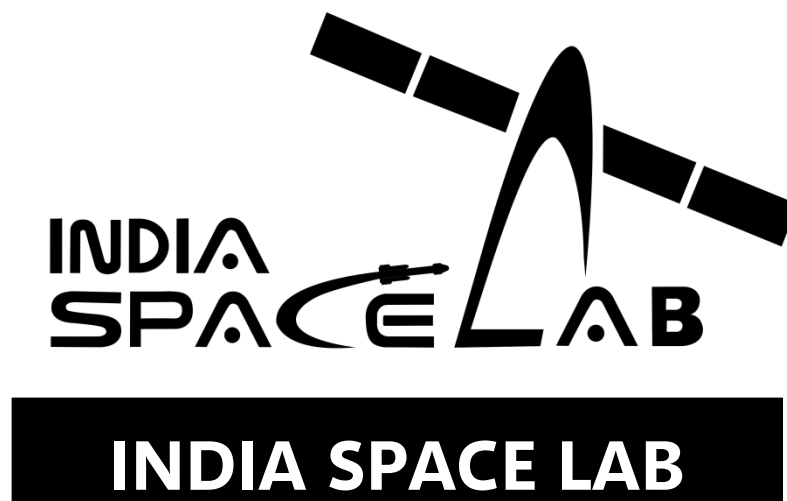


ISRO-Inspired CanSat Ground Control Software (GCS): Real-Time Telemetry Monitoring, Mission Operations, and Payload Recovery System



Project Report

01ST May to 15th June 2026

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ACKNOWLEDGEMENT

I express my sincere gratitude to India Space Lab for providing me with the valuable opportunity to undertake this project titled **“ISRO-Inspired CanSat Ground Control Software (GCS): Real-Time Telemetry Monitoring, Mission Operations, and Payload Recovery System.”** This project has been an enriching learning experience that enhanced my understanding of aerospace systems, telemetry processing, mission control operations, satellite communication concepts, and modern software development technologies.

I would like to extend my heartfelt thanks to the coordinators, mentors, and technical experts at India Space Lab for their continuous guidance, encouragement, and support throughout the internship period. Their expertise, constructive feedback, and willingness to share knowledge helped me gain practical insights into CanSat missions, telemetry monitoring, payload recovery systems, fault detection mechanisms, and aerospace mission operations. Their mentorship played a vital role in the successful completion of this project.

I am deeply grateful to my institution and respected faculty members for providing the academic foundation and motivation required to undertake this study. Their constant encouragement, valuable suggestions, and technical guidance helped me approach the project systematically and effectively. I also thank my classmates and peers for their support, discussions, and cooperation during various stages of the project.

I would like to acknowledge the contributions of the open-source development community and the organizations responsible for the technologies utilized in this project. Frameworks and libraries such as React.js, Vite, Tailwind CSS, Zustand, Three.js, Leaflet.js, and Chart.js provided the technological foundation necessary for developing a modern and interactive Ground Control Software platform. Their documentation, tutorials, and community support were valuable resources during the implementation process.

Special appreciation is extended to researchers, aerospace engineers, and software developers whose work in the fields of satellite communication, telemetry systems, mission control software, visualization technologies, and aerospace operations inspired the design and development of this project. Their innovations and contributions continue to advance the field of space technology and educational aerospace programs.

I am also thankful to the developers and contributors of mapping technologies, telemetry visualization tools, and simulation platforms that enabled the creation of realistic mission scenarios and recovery tracking capabilities. Their efforts helped provide practical exposure to real-world aerospace mission workflows and operational procedures.

Finally, I express my heartfelt gratitude to my family for their unwavering support, patience, encouragement, and motivation throughout the development of this project. Their constant belief in my abilities inspired me to overcome challenges and successfully complete this work.

I sincerely thank everyone who directly or indirectly contributed to the successful completion of this project and made this learning experience meaningful, informative, and rewarding.

ABSTRACT

CanSat missions provide an effective platform for simulating real-world satellite operations within a compact and cost-efficient environment. Ground Control Systems (GCS) play a critical role in monitoring telemetry, visualizing mission status, tracking payload location, and supporting recovery operations throughout the mission lifecycle.

This project presents the design and development of an ISRO-inspired CanSat Ground Control Software (GCS) capable of real-time telemetry monitoring, mission phase tracking, 3D attitude visualization, GNSS-based payload tracking, command uplink management, and fault diagnosis. The system is developed using modern web technologies including React, Vite, Tailwind CSS, Zustand, Three.js, Leaflet.js, and Chart.js.

The proposed system simulates an end-to-end CanSat mission consisting of eight operational phases: Pre-Launch, Launch, Ascent, Apogee, Separation, Descent, Recovery, and Mission Complete. A centralized simulation engine generates telemetry packets containing altitude, pressure, temperature, battery voltage, signal strength, GPS coordinates, and inertial sensor data. These packets are processed through a centralized state management architecture and displayed through interactive dashboards, real-time graphs, and 3D visualizations.

The Ground Control Software provides mission operators with situational awareness through dynamic telemetry analytics, payload video monitoring, GPS recovery mapping, and automated fault detection mechanisms. Historical mission data can be exported as CSV files and graphical reports for post-flight analysis.

The developed system demonstrates how modern web technologies can be utilized to create a scalable and responsive mission control environment suitable for CanSat missions, educational aerospace programs, and telemetry visualization research.

CHAPTER 1

INTRODUCTION

1.1 Background

The increasing demand for low-cost satellite missions has led to the development of CanSat systems, which are miniature satellites designed to fit within the volume of a standard beverage can. CanSat missions provide students and researchers with practical exposure to aerospace engineering concepts including telemetry acquisition, sensor integration, communication systems, mission planning, payload deployment, and recovery operations.

A Ground Control System (GCS) serves as the primary interface between the mission operators and the airborne CanSat payload. It enables real-time monitoring of mission parameters, visualization of flight status, communication with onboard systems, and analysis of collected telemetry data.

Modern aerospace missions require advanced monitoring solutions capable of processing large volumes of telemetry information while providing intuitive visualizations and rapid decision-making support. The development of web-based mission control systems offers several advantages including platform independence, scalability, accessibility, and ease of deployment.

1.2 Problem Statement

Traditional telemetry monitoring systems often rely on specialized desktop applications that may lack flexibility, advanced visualization capabilities, and modern user interfaces. Furthermore, educational CanSat missions require cost-effective solutions that can simulate professional aerospace mission operations while remaining accessible to students and researchers.

There is a need for a modern Ground Control Software capable of:

- Real-time telemetry visualization
- Mission phase tracking
- Payload location monitoring
- 3D attitude visualization
- Telemetry data logging
- Fault detection and diagnosis
- Recovery support operations

The proposed system addresses these requirements through an integrated web-based mission control environment.

1.3 Aim of the Project

The aim of this project is to design and develop an ISRO-inspired CanSat Ground Control Software capable of providing real-time telemetry monitoring, mission visualization, command management, and payload recovery support using modern web technologies.

1.4 Objectives

The primary objectives of the project are:

1. To develop a real-time telemetry monitoring dashboard.
2. To implement mission phase tracking and visualization.
3. To provide 3D orientation visualization using inertial sensor data.
4. To enable GPS-based payload tracking and recovery monitoring.
5. To implement telemetry analytics and graph visualization.
6. To support data export for post-mission analysis.
7. To develop a fault detection and diagnosis module.
8. To simulate a complete CanSat mission lifecycle.

1.5 Scope of the Project

The developed Ground Control Software supports the complete CanSat mission lifecycle from launch preparation to payload recovery. The system can be used for:

- CanSat competitions
- Aerospace education
- Telemetry visualization research
- Mission simulation environments
- Ground station operator training
- Embedded systems testing

The modular architecture also enables future integration with actual flight hardware and communication modules.

1.6 Overview of the Proposed System

The proposed system consists of multiple interconnected modules that work together to provide real-time mission monitoring and control.

The telemetry simulation engine generates mission data at one-second intervals and forwards telemetry packets to a centralized state management system. The Ground Control Software then distributes this information to various visualization modules including:

- Telemetry Dashboard
- Mission Sequence Tracker
- 3D Orientation Viewer
- GPS Recovery Map
- Payload Monitoring System
- Telemetry Analytics Engine
- Fault Detection and Diagnosis Module

The system continuously updates mission information and provides operators with situational awareness throughout all mission phases.

1.7 Mission Phases

The CanSat mission simulation consists of eight operational phases:

Phase 1: Pre-Launch Standby

System initialization and readiness verification.

Phase 2: Launch Profile

Launch sequence initiation and telemetry acquisition.

Phase 3: Ascent Gradient

Payload ascends while environmental parameters are monitored.

Phase 4: Apogee Detection

Maximum altitude is reached and mission status is updated.

Phase 5: Payload Separation

CanSat deployment and activation of recovery systems.

Phase 6: Parachute Descent

Controlled descent with continuous telemetry monitoring.

Phase 7: Recovery Beacon

GPS tracking and payload localization.

Phase 8: Mission Complete

Mission termination and final data logging.

1.8 Benefits of the System

The proposed Ground Control Software provides several advantages:

- Real-time telemetry visualization
- Interactive mission monitoring
- Enhanced situational awareness
- Improved recovery operations
- Historical data export capability
- Modern aerospace-inspired user interface
- Scalable architecture
- Cross-platform accessibility

1.9 Chapter Summary

This chapter introduced the concept of CanSat missions, discussed the need for a modern Ground Control System, presented the project objectives, and provided an overview of the proposed system architecture. The next chapter presents the literature survey and related work in telemetry monitoring systems, mission control software, and CanSat ground station technologies.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The increasing occurrence of climate-related disasters and environmental challenges has highlighted the importance of advanced technological systems for monitoring, prediction, decision-making, and disaster management. Modern technologies such as Artificial Intelligence (AI), Machine Learning (ML), Geographic Information Systems (GIS), Remote Sensing, Satellite Communication Systems, Internet of Things (IoT), and Unmanned Aerial Vehicles (UAVs) play a crucial role in collecting, processing, analyzing, and visualizing environmental data.

In the context of heat wave management, these technologies facilitate early warning generation, risk assessment, vulnerability mapping, real-time monitoring, and informed decision-making. This chapter reviews recent research studies related to satellite communication systems, GIS applications, AI-based monitoring systems, UAV platforms, and intelligent data processing technologies that can support effective heat wave management and climate resilience planning.

2.2 Literature Survey Table

Sl. No.	Authors	Year	Paper Title	Methodology	Key Findings
1	Zhan et al.	2021	Challenges and Solutions for the Satellite Tracking, Telemetry, and Command System	Satellite Communication Analysis	Advanced TT&C systems improve reliability and efficiency in satellite operations and data transmission.
2	Wang et al.	2018	Design of Reconfigurable Real-Time Telemetry Monitoring and Quantitative Management System for Remote Sensing Satellite in Orbit	Telemetry System Design	Real-time telemetry systems enhance monitoring and management of satellite operations.
3	Dao and Havlick	2025	Portraying the Geography of US Airspace with 3-Dimensional GIS-Based Analysis and Visualization	GIS-Based Visualization	GIS visualization improves spatial analysis and geographical decision-making capabilities.
4	Lin et al.	2023	Application of Data Visualization Interaction Technology in	Data Visualization Techniques	Interactive visualization significantly improves interpretation of

			Aerospace Data Processing		large-scale aerospace datasets.
5	Serief et al.	2023	Deep-Learning-Based System for Change Detection Onboard Earth Observation Small Satellites	Deep Learning and Remote Sensing	Deep learning improves automated change detection using satellite imagery.
6	Alghananim et al.	2025	Machine Learning-Based Fault Detection and Exclusion for GNSS Pseudorange Measurements	Machine Learning	Machine learning algorithms effectively identify faults and improve navigation system reliability.
7	Van Gompel et al.	2022	Satellite-Based Fault Diagnosis of Photovoltaic Systems Using Recurrent Neural Networks	Neural Network-Based Diagnostics	Recurrent Neural Networks provide accurate fault diagnosis and predictive monitoring capabilities.
8	Hilton et al.	2024	Intelligent Mission Planning for Autonomous Distributed Satellite Systems	Intelligent Decision Systems	AI-based planning systems improve mission efficiency and autonomous decision-making.
9	Wang et al.	2025	Toward Intelligent Space-Air-Ground Integrated Network	Integrated Network Architecture	Integrated communication networks support seamless data sharing and intelligent monitoring.
10	Kotovs and Zacepins	2023	GIS-Based Interactive Map to Improve Scheduling Beekeeping Activities	GIS-Based Decision Support	GIS platforms improve resource management and operational planning.
11	Poma et al.	2024	Open-Source Web-Based Ground Control Station for Long-Range Inspection with Multiple UAVs	UAV Monitoring System	Web-based UAV control systems improve remote monitoring and operational efficiency.
12	Raj, Sinha and Gautam	2024	Designing and Development of Can-Sized Satellite (CanSat): Learnings, Challenges, and Implementations	Small Satellite Development	CanSat systems provide practical experience in satellite communication,

					sensing, and data acquisition.
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2.3 Review of Existing Studies

Paper 1: Satellite Tracking, Telemetry, and Command Systems (Zhan et al., 2021)

The authors examined the challenges associated with satellite tracking, telemetry, and command systems and proposed solutions for improving communication reliability and operational efficiency. The study highlighted the importance of robust communication infrastructure for real-time environmental monitoring.

Research Gap: Limited focus on integration with disaster management applications.

Paper 2: Real-Time Telemetry Monitoring Systems (Wang et al., 2018)

This research proposed a reconfigurable real-time telemetry monitoring framework for remote sensing satellites. The system improved monitoring accuracy and data management efficiency.

Research Gap: Limited application toward environmental hazard monitoring and climate resilience systems.

Paper 3: GIS-Based Spatial Visualization (Dao and Havlick, 2025)

The study demonstrated the effectiveness of three-dimensional GIS visualization for spatial data analysis and decision-making. GIS tools provided enhanced visualization capabilities for complex geographical datasets.

Research Gap: Limited implementation for heat wave vulnerability assessment and disaster preparedness.

Paper 4: Aerospace Data Visualization Technology (Lin et al., 2023)

The researchers explored advanced visualization techniques for aerospace data processing. Interactive systems significantly improved user understanding and interpretation of large datasets.

Research Gap: Limited use in climate-related risk communication systems.

Paper 5: Deep Learning-Based Change Detection (Serief et al., 2023)

This work developed a deep-learning framework for detecting environmental changes using satellite imagery. The model demonstrated high accuracy and automation capabilities.

Research Gap: Limited application toward urban heat island and heat stress monitoring.

Paper 6: Machine Learning-Based Fault Detection (Alghananim et al., 2025)

The study utilized machine learning techniques to identify faults in satellite navigation systems. Results demonstrated improved system reliability and decision-making accuracy.

Research Gap: Limited adaptation for climate monitoring and disaster forecasting systems.

Paper 7: Neural Network-Based Fault Diagnosis (Van Gompel et al., 2022)

The authors proposed recurrent neural network models for diagnosing faults in photovoltaic systems using satellite-derived information. The approach improved predictive maintenance capabilities.

Research Gap: Limited focus on environmental disaster management applications.

Paper 8: Intelligent Mission Planning Systems (Hilton et al., 2024)

The study introduced autonomous mission planning techniques for distributed satellite systems. Intelligent planning significantly improved operational performance and resource allocation.

Research Gap: Further research is required for integrating such systems into environmental monitoring platforms.

Paper 9: Intelligent Space-Air-Ground Networks (Wang et al., 2025)

This paper proposed an integrated communication architecture connecting space, aerial, and terrestrial networks. The architecture enables efficient data sharing and large-scale monitoring capabilities.

Research Gap: Limited evaluation for heat wave monitoring and disaster response applications.

Paper 10: GIS-Based Interactive Mapping Systems (Kotovs and Zacepins, 2023)

The authors developed GIS-based decision support systems for resource management. The study demonstrated the effectiveness of GIS tools in improving planning and operational efficiency.

Research Gap: Limited focus on climate vulnerability mapping.

Paper 11: UAV Ground Control Systems (Poma et al., 2024)

This study proposed an open-source web-based control station for UAV operations. The system enhanced monitoring efficiency and real-time decision support.

Research Gap: Further investigation is needed for heat wave monitoring and emergency response applications.

Paper 12: CanSat Development and Implementation (Raj et al., 2024)

The researchers presented the design and development of a CanSat platform, highlighting practical challenges and learning experiences in satellite-based sensing and communication systems.

Research Gap: Limited exploration of environmental disaster monitoring applications.

2.4 Comparative Analysis

The reviewed studies collectively demonstrate the growing importance of AI, GIS, satellite communication systems, telemetry technologies, UAV platforms, remote sensing, and intelligent network architectures. These technologies contribute significantly to environmental monitoring, spatial analysis, predictive modeling, and decision support systems. GIS and remote sensing technologies enable vulnerability mapping and environmental assessment, while AI and machine learning improve prediction accuracy and automation capabilities. UAVs and satellite systems further enhance monitoring coverage and data collection efficiency.

2.5 Research Gap Identification

The literature review identifies the following research gaps:

1. Limited integration of AI, GIS, Remote Sensing, and UAV technologies into a unified heat wave management framework.
2. Insufficient focus on heat wave prediction and vulnerability assessment using satellite-based systems.
3. Limited use of intelligent communication networks for disaster preparedness.
4. Lack of real-time environmental monitoring platforms combining multiple technologies.
5. Need for cost-effective and scalable technological solutions for climate resilience.
6. Limited integration of satellite, aerial, and ground-based monitoring systems.

2.6 Proposed Research Direction

The present study proposes a comprehensive framework integrating:

- Artificial Intelligence-Based Forecasting
- GIS-Based Heat Vulnerability Mapping
- Remote Sensing Technologies
- IoT-Based Environmental Monitoring
- Mobile Alert Systems
- UAV-Assisted Monitoring
- Smart Decision Support Systems

The proposed framework aims to enhance preparedness, response, recovery, and long-term resilience against heat wave disasters.

2.7 Conclusion

The literature review demonstrates that advanced technologies such as AI, GIS, Remote Sensing, Satellite Communication Systems, UAVs, and Intelligent Monitoring Networks have significant potential to improve disaster management and environmental monitoring. Integrating these technologies into heat wave management systems can enhance forecasting accuracy, vulnerability assessment, risk communication, and decision-making capabilities. Future research should focus on developing unified and scalable technology-driven frameworks for climate resilience and disaster preparedness.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

The CanSat Ground Control Software (GCS) is designed as a web-based mission operations platform that provides real-time telemetry monitoring, mission phase tracking, payload recovery support, fault diagnosis, and aerospace visualization. The system follows a modular architecture where telemetry data is generated, processed, stored, and visualized through independent but interconnected software components.

The architecture ensures scalability, maintainability, and efficient real-time performance while providing operators with comprehensive mission awareness.

3.2 System Overview

The proposed Ground Control System consists of:

1. Flight Simulation Engine
2. Telemetry Processing Engine
3. State Management Layer
4. Visualization Layer
5. Mapping System
6. Command Management Module
7. Fault Detection System
8. Export and Reporting Module

The entire system is developed using modern JavaScript technologies and operates within a web browser environment.

3.3 Overall System Architecture

The architecture follows a centralized telemetry processing model.

Figure 3.1 – System Architecture Diagram

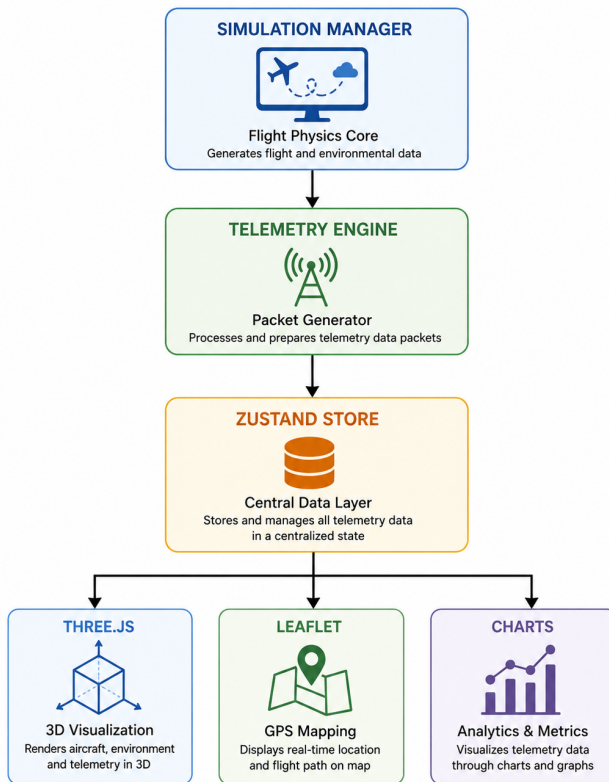


Fig. 3.1 System Architecture Diagram

The telemetry data generated by the simulation manager is transmitted to the telemetry engine, which processes the information and updates the centralized Zustand store. All user interface components retrieve information from the store and update automatically.

3.4 Data Flow Analysis

The telemetry workflow follows the sequence below:

Step 1

The simulation engine generates:

- Altitude
- Pressure
- Temperature
- Voltage
- RSSI
- GPS Coordinates
- Roll
- Pitch
- Yaw

values at one-second intervals.

Step 2

Telemetry packets are transmitted to the telemetry engine.

Step 3

The telemetry engine validates and processes the packets.

Step 4

Processed telemetry is stored inside the Zustand state store.

Step 5

UI modules subscribe to the store and update in real-time.

Step 6

Operators monitor mission progress through:

- Dashboard Panels
- 3D Viewer
- Recovery Map
- Graph Analytics
- Mission Tracker

Step 7

Historical telemetry can be exported for post-flight analysis.

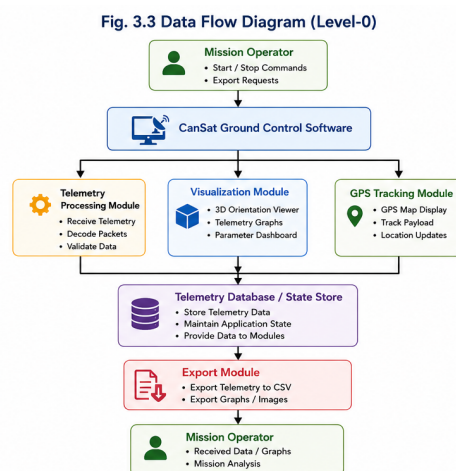


Figure 3.3 illustrates the data flow within the proposed CanSat Ground Control Software. Telemetry packets generated by the simulation engine are processed by the telemetry engine and stored in the centralized state store. The processed data is then distributed to various visualization modules including the 3D orientation viewer, GPS tracking module, fault detection system, and telemetry analytics dashboard. The mission operator interacts with the system through control commands and can export telemetry data and graphs for post-mission analysis.

Fig. 3.3 Data Flow Diagram of the Proposed CanSat Ground Control Software

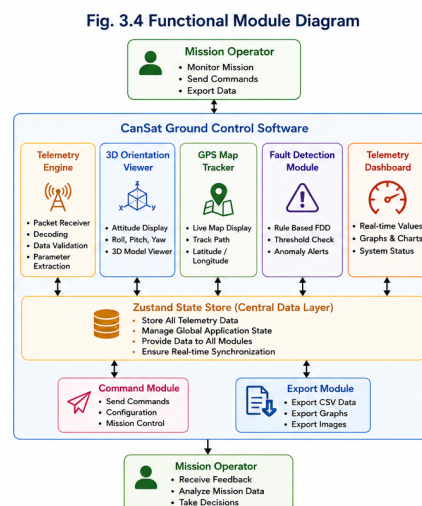


Figure 3.4 shows the functional modules of the CanSat Ground Control Software. Each module performs a specific function such as telemetry processing, 3D orientation visualization, GPS tracking, fault detection, telemetry dashboard, command handling, and data export. The Zustand state store acts as the central data layer that maintains application state and ensures seamless data sharing among all modules.

Fig. 3.4 Functional Module Diagram of the Proposed CanSat Ground Control Software

Fig 3.2 Data Flow Diagram

3.5 Functional Modules

The system is divided into several modules.

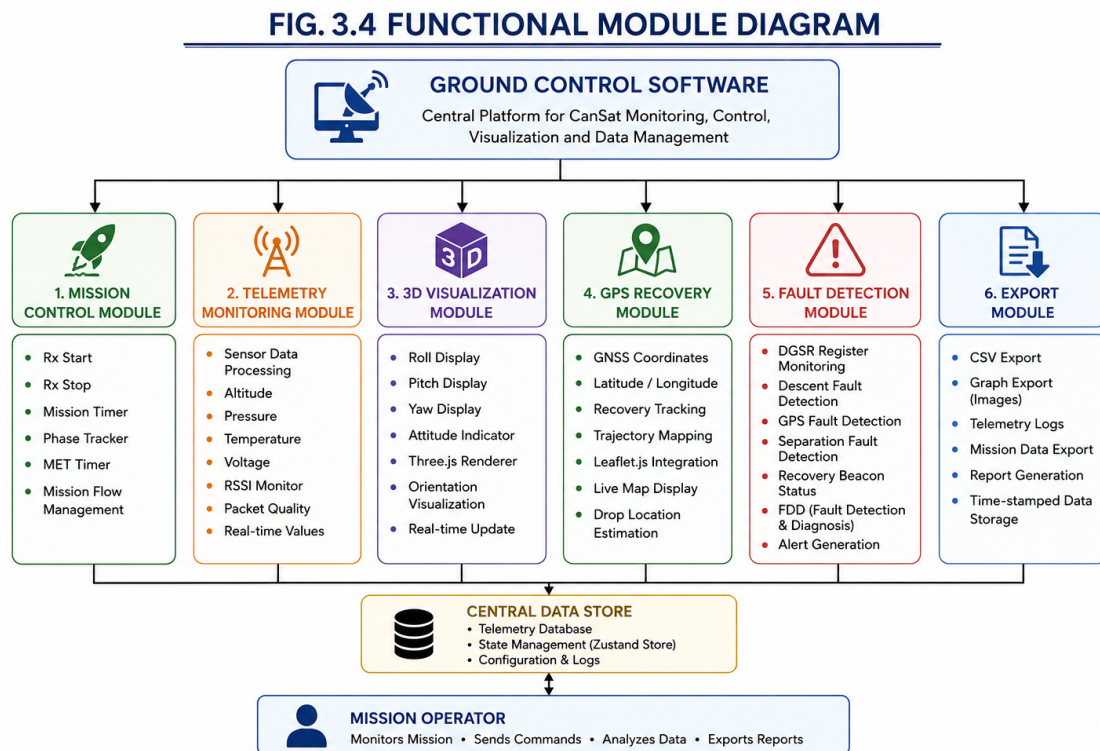


Fig. 3.4 Functional Module Diagram of the Proposed CanSat Ground Control Software

Fig. 3.3 Functional Module Diagram of the Proposed CanSat Ground Control Software

3.5.1 Mission Control Module

This module manages:

- Mission Start
- Mission Stop
- Time Synchronization
- Mission Status Updates

Functions:

- Controls telemetry reception
- Tracks Mission Elapsed Time (MET)
- Maintains mission states

3.5.2 Telemetry Monitoring Module

This module continuously displays:

- Altitude
- Pressure
- Temperature
- Voltage
- RSSI
- GPS Coordinates

Features:

- Live updates
- Dynamic indicators
- Warning generation

Benefits:

- Continuous mission monitoring
- Improved operator awareness

3.5.3 3D Attitude Visualization Module

This module is implemented using Three.js.

Functions:

- Displays Roll
- Displays Pitch
- Displays Yaw

Features:

- Real-time orientation updates
- Smooth animation
- Aerospace-style visualization

Benefits:

- Easy interpretation of INS data
- Better mission awareness

3.5.4 Flight Sequence Tracking Module

Tracks mission phases:

1. Pre-Launch Standby
2. Launch Profile
3. Ascent Gradient
4. Apogee Detection
5. Payload Separation
6. Parachute Descent
7. Recovery Beacon

8. Mission Complete

Benefits:

- Mission progress visualization
- Operational status awareness

3.5.5 Telemetry Analytics Module

Implemented using Chart.js.

Monitors:

- Altitude Trends
- Descent Velocity
- Temperature Changes
- Pressure Changes
- Voltage Behavior
- RSSI Variation

Features:

- Real-time graphs
- Historical data storage
- Export capability

3.5.6 GPS Recovery Module

Implemented using Leaflet.js.

Functions:

- Real-time location display
- Recovery path generation
- Payload tracking

Features:

- Dark map theme
- Radar marker
- Trajectory visualization

Benefits:

- Faster payload recovery
- Improved field operations

3.5.7 Command Uplink Module

Provides command transmission capability.

Commands include:

- Manual Separation
- Emergency Parachute
- GPS Recalibration
- Beacon Activation
- CPU Bus Activation

Features:

- Safety Interlock System
- Command Validation
- Event Logging

3.5.8 Fault Detection and Diagnosis Module

Monitors mission health through DGSR registers.

Registers:

Register Description

D	Descent Velocity Fault
G	GPS Fault
S	Separation Failure
R	Recovery Beacon Status

Functions:

- Fault Identification
- Alert Generation
- Health Monitoring

Benefits:

- Increased mission safety
- Early fault detection

3.6 Software Requirements

The project uses the following software technologies.

Component	Technology
Frontend Framework	React.js

Build Tool	Vite
Styling Framework	Tailwind CSS
State Management	Zustand
3D Visualization	Three.js
Mapping System	Leaflet.js
Analytics	Chart.js
Programming Language	JavaScript
Package Manager	npm
Browser	Chrome/Edge/Firefox

Table 3.1 Software Requirements

3.7 Hardware Requirements

Minimum System Requirements:

Component	Specification
Processor	Intel i5 or Higher
RAM	8 GB
Storage	10 GB Free Space
Graphics	Integrated GPU
Operating System	Windows/Linux/macOS
Internet	Optional

Table 3.2 Hardware Requirements

3.8 Use Case Analysis

Primary Actor

Mission Operator

Use Case Diagram – CanSat Ground Control Software

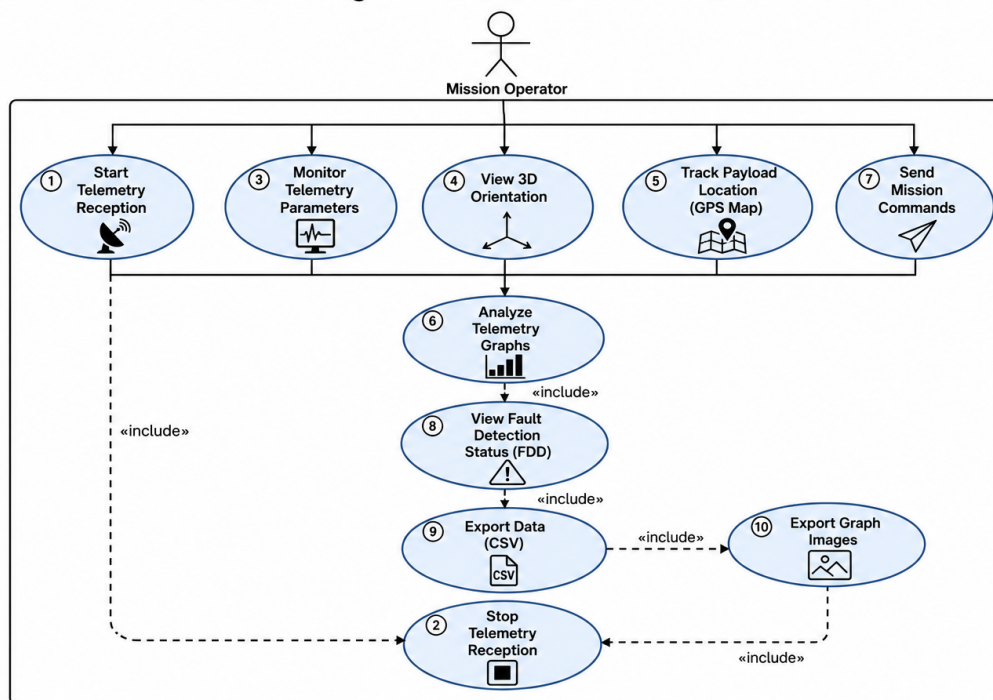


Fig. 3.2 Use Case Diagram of the Proposed CanSat Ground Control Software

Fig 3.4 Use Case Diagram

Use Cases

- Start Telemetry Reception
- Stop Telemetry Reception
- Monitor Telemetry
- View 3D Orientation
- Track Payload
- Analyze Graphs
- Send Commands
- Export Data
- Review Mission Logs

The Mission Operator interacts with all major modules through the dashboard interface.

3.9 Advantages of Proposed Design

The proposed architecture provides:

- Real-time performance
- Modular implementation
- Scalability
- Easy maintenance
- Cross-platform operation
- Interactive visualization
- Integrated recovery support

- Aerospace-inspired user experience

3.10 Chapter Summary

This chapter described the architecture, modules, workflow, software stack, and operational design of the proposed CanSat Ground Control Software. The modular architecture enables efficient telemetry processing, visualization, and mission management. The next chapter presents the implementation methodology and detailed development process of the system.

CHAPTER 4

SYSTEM IMPLEMENTATION AND METHODOLOGY

4.1 Introduction

The CanSat Ground Control Software (GCS) was developed using modern web technologies to provide a responsive, scalable, and real-time mission monitoring environment. The implementation follows a modular architecture where telemetry generation, state management, visualization, analytics, and recovery tracking are handled by independent software components.

The system simulates a complete CanSat mission and continuously generates telemetry data for visualization, analysis, and recovery operations.

4.2 Development Methodology

The project follows an iterative development approach consisting of:

Phase 1: Requirement Analysis

Requirements were identified based on:

- CanSat mission objectives
- Aerospace mission control systems
- Telemetry monitoring requirements
- Payload recovery requirements
- Real-time visualization needs

Phase 2: System Design

The architecture was designed to support:

- Real-time telemetry processing
- Centralized state management
- Modular user interface
- Scalable visualization system

Phase 3: Development

Implementation involved:

- Dashboard development
- Telemetry engine creation

- Mission simulation development
- GPS tracking implementation
- Fault detection integration

Phase 4: Testing

Testing was performed for:

- Telemetry accuracy
- Mission phase transitions
- Graph generation
- GPS updates
- Export functionality

4.3 Frontend Implementation

The user interface was developed using React.js.

React provides:

- Component-based architecture
- Efficient rendering
- State synchronization
- Reusable UI modules

The dashboard consists of several independent components:

Mission Header
Telemetry Panel
Mission Tracker
3D Viewer
Analytics Dashboard
Recovery Map
Command Panel
Fault Detection Panel

Each component subscribes to the centralized telemetry state and updates automatically when new telemetry packets arrive.

4.4 State Management Using Zustand

The project uses Zustand as the centralized state management system.

The store maintains:

- Mission status
- Telemetry packets
- GPS information
- Orientation data
- Sensor readings

- Historical records

Responsibilities

- Store mission state
- Synchronize UI updates
- Manage telemetry history
- Handle mission events

Advantages:

- Lightweight architecture
- Fast performance
- Reduced complexity
- Efficient rendering

4.5 Simulation Manager Implementation

The simulation manager acts as the flight physics engine.

Functions:

- Generate flight trajectories
- Simulate atmospheric conditions
- Create mission events
- Model payload descent

Generated Parameters:

Environmental Data

- Temperature
- Pressure

Flight Data

- Altitude
- Vertical Velocity

Navigation Data

- Latitude
- Longitude

Attitude Data

- Roll
- Pitch
- Yaw

The simulation engine updates these parameters every second to mimic real mission behavior.

4.6 Telemetry Engine Implementation

The telemetry engine acts as the communication layer between the simulation manager and the dashboard.



Fig. 4.1 Pre-Launch Dashboard

Responsibilities:

Packet Generation

Creates telemetry packets containing:

Altitude
Pressure
Temperature
Voltage
RSSI
Latitude
Longitude
Roll
Pitch
Yaw
Mission Phase

Packet Processing

- Validates incoming values
- Updates mission state
- Stores historical records

Data Distribution

After processing, telemetry packets are forwarded to the Zustand store for visualization.

4.7 Mission Phase Automation

The telemetry engine automatically transitions between mission phases.

Mission Phases

Phase 1

Pre-Launch Standby

System initialization and readiness verification.

Phase 2

Launch Profile

Telemetry transmission begins.

Phase 3

Ascent Gradient

Altitude increases continuously.

Phase 4

Apogee Detection

Maximum altitude reached.

Phase 5

Payload Separation

CanSat separates from carrier vehicle.

Phase 6

Parachute Descent

Controlled descent begins.

Phase 7

Recovery Beacon

GPS tracking activated.

Phase 8

Mission Complete

Flight terminated and logs finalized.

The mission tracker updates automatically based on current phase.

4.8 Three.js Attitude Visualization

The project uses Three.js to implement the 3D orientation viewer.

Purpose

Visualize:

- Roll
- Pitch
- Yaw

in real time.

Features

- Cylindrical CanSat model
- Smooth animation
- Real-time updates
- Aerospace visualization

Working Principle

Orientation values received from telemetry packets are converted into rotation vectors and applied to the Three.js model.

Linear interpolation (LERP) is used to achieve smooth rotational transitions.

Benefits:

- Easy interpretation of INS data
- Improved situational awareness

4.9 Telemetry Analytics Using Chart.js

Chart.js is used to generate real-time analytical graphs.

Graph 1

Altitude vs Vertical Velocity

Purpose:

- Monitor ascent profile
- Analyze descent performance

Graph 2

Temperature vs Pressure

Purpose:

- Observe atmospheric variations
- Validate environmental trends

Graph 3

Voltage vs RSSI

Purpose:

- Monitor communication quality
- Track battery discharge

Features

- Dynamic updates
- Historical storage
- Export support

4.10 GPS Tracking and Recovery System

The recovery module is implemented using Leaflet.js.



Fig. 4.2 Ascent Phase Dashboard

Functions

- Display payload position
- Track flight path
- Assist recovery operations

Features

- Real-time coordinates
- Dark theme mapping
- Radar-style marker
- Polyline trajectory

The module continuously updates payload location using GNSS telemetry.

Benefits:

- Faster payload retrieval
- Improved mission efficiency

4.11 Command Uplink System

The command panel allows operators to send mission commands.

Available Commands

1. Manual Separation
2. Emergency Parachute
3. GPS Recalibration
4. Beacon Activation
5. Redundant CPU Bus Activation

Safety Features

- Interlock protection
- Command validation
- Event logging

Each command generates:

SUCCESS
PENDING
FAILED

status responses.

4.12 Fault Detection and Diagnosis System

The Fault Detection and Diagnosis (FDD) module continuously monitors mission health.

DGSR Registers

Register	Description
D	Descent Velocity Fault
G	GPS Boundary Fault
S	Separation Failure
R	Recovery Beacon Active

Table 4.1 DGSR Fault Register Description

Fault Logic

D Fault

Triggered when:

Descent Velocity > 12 m/s

G Fault

Triggered when:

GPS Coordinates Invalid

S Fault

Triggered when:

Payload Separation Fails

R Fault

Triggered when:

Recovery Beacon Activated

Benefits:

- Early anomaly detection
- Mission safety improvement
- Reduced operational risk

4.13 CSV Export Module

The system provides telemetry export functionality.

Exported Data

- Mission Time
- Altitude
- Pressure
- Temperature
- Voltage
- RSSI
- GPS Coordinates

The exported CSV file can be used for:

- Post-flight analysis
- Mission reporting
- Statistical evaluation

4.14 Graph Export Module

Graph images can be exported for documentation purposes.

Supported Graphs:

- Altitude vs Velocity
- Temperature vs Pressure
- Voltage vs RSSI

Benefits:

- Easy report preparation
- Mission performance analysis
- Presentation support

4.15 User Interface Design

The dashboard follows an aerospace-inspired design philosophy.

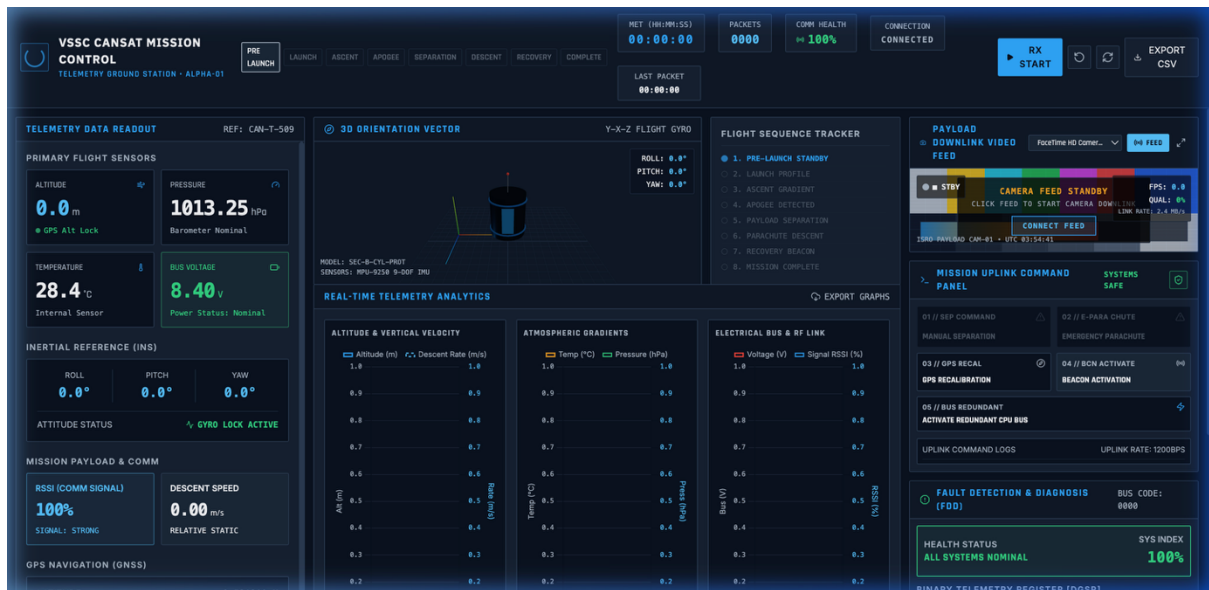


Fig. 4.3 Descent Phase Dashboard

Design Features:

- Dark theme interface
- High-contrast telemetry panels
- Responsive layout
- Mission-oriented workflow
- Minimal operator distraction

The interface is optimized for continuous monitoring during active missions.

4.16 Chapter Summary



Fig. 4.4 Mission Complete Dashboard

This chapter described the implementation of the CanSat Ground Control Software, including the telemetry engine, simulation manager, centralized state management, 3D visualization system, GPS tracking module, analytics engine, fault detection mechanism, and export services. The developed system successfully integrates all major mission control functionalities into a single web-based platform. The next chapter presents testing procedures, mission execution results, telemetry analysis, and performance evaluation.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the results obtained from the execution of the CanSat Ground Control Software during a complete simulated mission. The mission was executed through all predefined operational phases, including pre-launch, ascent, apogee detection, payload separation, descent, recovery, and mission completion.

The system successfully demonstrated real-time telemetry monitoring, mission phase tracking, GPS recovery mapping, 3D orientation visualization, fault monitoring, command execution, and data export functionalities.

5.2 Mission Execution Overview

The mission simulation progressed through eight operational phases:

Phase	Description
Pre-Launch	System initialization
Launch	Mission start
Ascent	Altitude increase
Apogee	Maximum altitude reached
Separation	Payload deployment
Descent	Controlled parachute descent
Recovery	GPS recovery phase
Complete	Mission termination

Table 5.1 Mission Phase Summary

The mission lifecycle was executed successfully without interruption.



Fig. 5.1 Complete Ground Control Dashboard during active mission operations.

5.3 Telemetry Performance Analysis

The telemetry engine generated packets at a frequency of approximately one packet per second.

Parameters Monitored

- Altitude
- Pressure
- Temperature
- Voltage
- RSSI

- Latitude
- Longitude
- Roll
- Pitch
- Yaw
- Descent Rate

The dashboard displayed all parameters in real time with minimal latency.

Observations

- Stable telemetry reception throughout the mission.
- Continuous packet updates without data loss.
- Successful synchronization between telemetry engine and dashboard modules.
- Accurate mission phase transitions.

5.4 Altitude Analysis

The altitude profile demonstrated the expected CanSat mission trajectory.

Observed Behavior

Pre-Launch

Altitude remained near ground level.

Launch Phase

Rapid altitude increase observed.

Ascent Phase

Continuous climb toward mission altitude.

Apogee

Maximum altitude approximately 1000 meters.

Descent

Controlled reduction in altitude due to parachute deployment.

Recovery

Altitude stabilized near ground level.

Result

The altitude graph accurately represented the complete mission profile from launch to landing.



Fig. 5.2 Exported telemetry graphs showing altitude, velocity, pressure, temperature, voltage, and RSSI trends.

5.5 Atmospheric Data Analysis

Pressure Variation

Pressure decreased as altitude increased.

This behavior follows the standard atmospheric model.

Observations:

- Maximum pressure observed near ground level.
- Minimum pressure observed near apogee.
- Pressure increased again during descent.

Temperature Variation

Temperature followed atmospheric lapse-rate behavior.

Observations:

- Higher temperatures near ground level.
- Gradual decrease during ascent.
- Temperature recovery during descent.

The generated data closely resembled real atmospheric conditions.

5.6 Battery Performance Analysis

Battery voltage was continuously monitored throughout the mission.

Observations

- Initial voltage approximately 8.4V.
- Gradual discharge observed during flight.
- Stable power availability throughout all mission phases.
- No critical low-voltage conditions detected.

Interpretation

The power subsystem maintained sufficient energy reserves for mission completion.

5.7 Communication System Analysis

Signal quality was evaluated using RSSI values.

Observations

- Strong signal strength during pre-launch.
- Minor fluctuations during ascent and descent.
- No communication loss detected.
- Telemetry stream remained active throughout the mission.

Result

Communication reliability remained within acceptable operational limits.

5.8 GPS Tracking Analysis

The GPS tracking system successfully recorded payload movement.

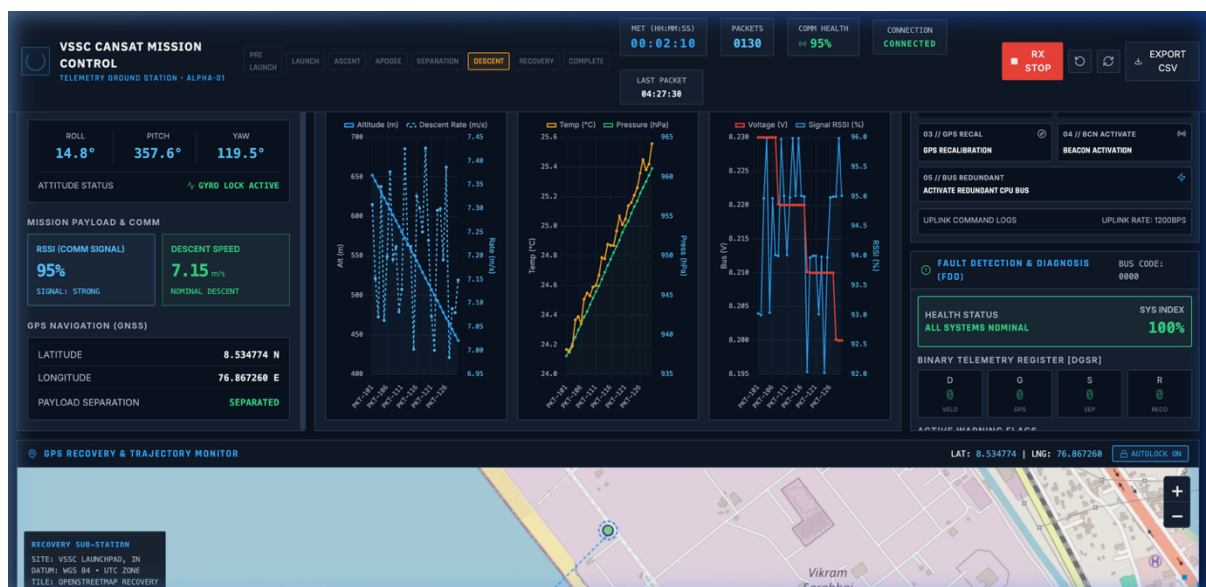


Fig. 5.3 Real-time GPS tracking and trajectory visualization.

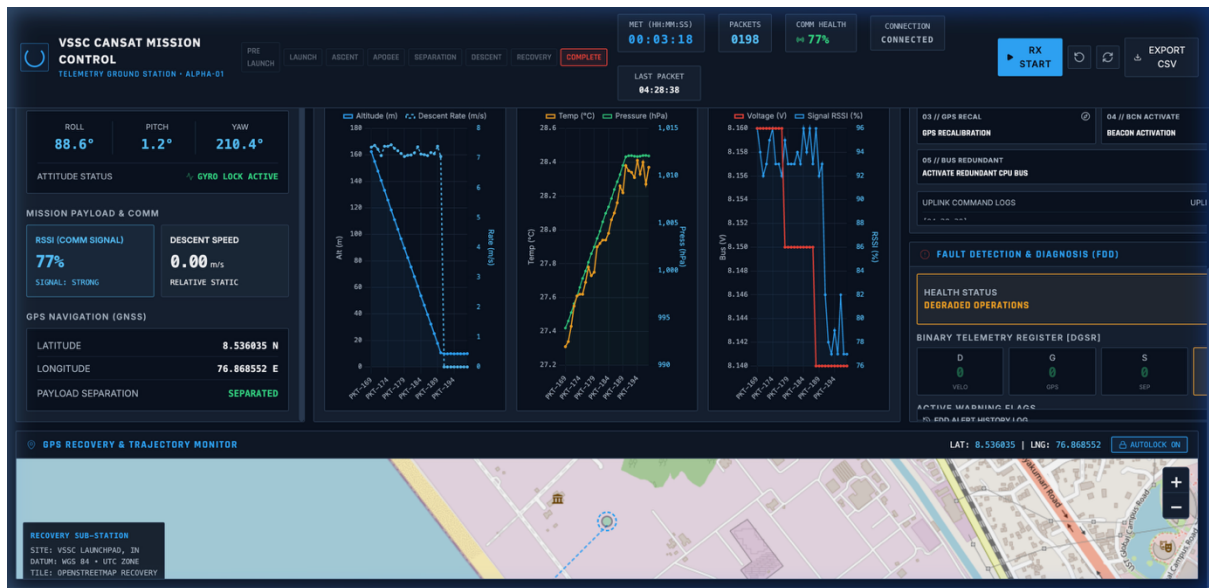


Fig. 5.4 Complete payload trajectory and recovery location.

Features Verified

- Real-time coordinate updates.
- Dynamic payload marker movement.
- Continuous trajectory generation.
- Recovery path visualization.

Recovery Support

The final payload location was successfully recorded, demonstrating the effectiveness of the recovery subsystem.

Result

GPS tracking functionality operated as expected and provided accurate recovery assistance.

5.9 3D Orientation Analysis

The Three.js visualization module successfully displayed payload orientation.

Parameters Visualized

- Roll
- Pitch
- Yaw

Observations

- Smooth rotational animation.
- Real-time synchronization with telemetry.

- Stable rendering performance.

Result

The orientation viewer provided an intuitive representation of inertial sensor data.

5.10 Command Uplink Performance

The command module was tested using simulated mission commands.

Commands Tested

1. Payload Separation
2. Emergency Parachute
3. GPS Recalibration
4. Beacon Activation
5. Redundant System Activation

Observations

- Commands executed successfully.
- Safety interlock prevented unauthorized operations.
- Event logging functioned correctly.

Result

The command subsystem operated reliably under test conditions.

5.11 Fault Detection and Diagnosis Results

The DGSR fault monitoring system continuously evaluated mission health.

Monitored Faults

Fault Code	Description
D	Descent Rate Fault
G	GPS Fault
S	Separation Fault
R	Recovery Beacon Status

Table 5.2 Fault Monitoring Summary

Observations

- Fault monitoring remained active throughout mission execution.
- Fault indicators updated dynamically.
- Health monitoring system responded correctly to simulated events.

Result

The FDD subsystem successfully identified mission status changes and potential anomalies.

5.12 Export Functionality Verification

The system supports:

CSV Export

Verified Features:

- Complete telemetry history exported.
- Mission timeline preserved.
- Sensor data stored correctly.



Fig. 5.5 CSV telemetry export operation.



Fig. 5.6 Telemetry graph export operation.

Graph Export

Verified Features:

- Telemetry charts exported successfully.
- High-resolution image generation.
- Suitable for documentation and analysis.

Result

Data export functionality met all project requirements.

5.13 User Interface Evaluation

The aerospace-inspired dashboard was evaluated based on usability and situational awareness.

Positive Features

- Clear telemetry presentation.
- Intuitive navigation.
- Consistent color coding.
- Responsive layout.
- Real-time updates.

Operator Benefits

- Faster decision making.
- Improved mission monitoring.
- Reduced information overload.

5.14 Telemetry CSV Export Output

The Ground Control Software provides a telemetry export feature that enables mission operators to download mission data in CSV format for post-mission analysis. The exported file contains packet number, timestamp, mission time, altitude, pressure, temperature, voltage, RSSI, GPS coordinates, attitude information, descent rate, and mission phase data.

Packet Num	Timestamp	Mission Time	Altitude (m)	Pressure (hPa)	Temp (°C)	Voltage (V)	RSSI (%)	Latitude	Longitude	Roll (°)	Pitch (°)	Yaw (°)	Descent Rate	Mission Phase
1	2026-06-07T00:00:01	0:00:01	10.14	1012.05	28.3	8.4	100	8.531204	76.864195	0.03	359.84	119.88	-0.01	PRE_LAUNCH
2	2026-06-07T00:00:02	0:00:02	10.02	1012.08	28.4	8.4	100	8.531198	76.864205	0.08	359.58	120.12	-0.01	PRE_LAUNCH
3	2026-06-07T00:00:03	0:00:03	9.81	1012.09	28.4	8.39	95	8.531201	76.864199	0.21	359.55	120.19	0.01	PRE_LAUNCH
4	2026-06-07T00:00:04	0:00:04	34.72	1009.01	28.1	8.39	95	8.53122	76.864211	43.05	5.21	122.22	-24.91	LAUNCH
5	2026-06-07T00:00:05	0:00:05	59.94	1006.12	28	8.39	100	8.53124	76.864219	86.56	5.5	124.02	-25.22	LAUNCH
6	2026-06-07T00:00:06	0:00:06	84.22	1003.16	27.8	8.39	100	8.531261	76.86423	131.12	5.67	125.77	-24.28	LAUNCH
7	2026-06-07T00:00:07	0:00:07	109.14	1000.28	27.7	8.39	96	8.531283	76.864242	174.26	5.85	127.79	-24.93	LAUNCH
8	2026-06-07T00:00:08	0:00:08	149.54	995.5	27.4	8.38	100	8.531324	76.864271	203.9	7.09	130.58	-40.4	ASCENT
9	2026-06-07T00:00:09	0:00:09	188.38	990.92	27.1	8.38	100	8.531368	76.864299	234.35	8.66	133.43	-38.84	ASCENT
10	2026-06-07T00:00:10	0:00:10	226.05	986.3	26.9	8.38	100	8.531406	76.864333	264.2	8.19	136.71	-37.67	ASCENT
11	2026-06-07T00:00:11	0:00:11	262.04	982.23	26.6	8.38	98	8.531451	76.864359	293.26	8.59	139.48	-35.99	ASCENT
12	2026-06-07T00:00:12	0:00:12	296.98	978.16	26.4	8.38	96	8.531492	76.864385	323.58	7.24	142.39	-34.94	ASCENT
13	2026-06-07T00:00:13	0:00:13	329.68	974.24	26.3	8.37	97	8.531527	76.864414	352.91	8.05	144.98	-32.69	ASCENT
14	2026-06-07T00:00:14	0:00:14	361.2	970.64	26.1	8.37	100	8.531569	76.864446	22.2	8.78	148.36	-31.53	ASCENT
15	2026-06-07T00:00:15	0:00:15	391.53	967.09	25.9	8.37	97	8.531606	76.864479	51.26	7.29	151.21	-30.32	ASCENT
16	2026-06-07T00:00:16	0:00:16	420.54	963.69	25.6	8.37	100	8.531649	76.864508	80.31	8.24	153.99	-29.01	ASCENT
17	2026-06-07T00:00:17	0:00:17	449.16	960.35	25.5	8.37	100	8.531693	76.864538	110.65	7.86	156.67	-28.62	ASCENT
18	2026-06-07T00:00:18	0:00:18	475.76	957.41	25.3	8.36	99	8.531729	76.864564	139.96	8.25	160.13	-26.6	ASCENT
19	2026-06-07T00:00:19	0:00:19	501.92	954.45	25.1	8.36	96	8.531764	76.864596	169.34	7.35	163.56	-26.16	ASCENT
20	2026-06-07T00:00:20	0:00:20	526.47	951.63	25	8.36	96	8.531803	76.864628	198.91	8.6	166.65	-24.55	ASCENT
21	2026-06-07T00:00:21	0:00:21	550.25	948.86	24.8	8.36	98	8.531843	76.86466	229.62	7.76	169.26	-23.78	ASCENT
22	2026-06-07T00:00:22	0:00:22	572.93	946.3	24.7	8.36	98	8.531881	76.864687	259.95	7.32	172.14	-22.69	ASCENT
23	2026-06-07T00:00:23	0:00:23	595	943.84	24.5	8.35	96	8.531923	76.864718	288.96	7.07	175.3	-22.07	ASCENT
24	2026-06-07T00:00:24	0:00:24	616	941.42	24.3	8.35	99	8.531964	76.864746	319.58	8.57	178.49	-20.99	ASCENT
25	2026-06-07T00:00:25	0:00:25	635.95	939.11	24.3	8.35	100	8.532001	76.864778	349.95	8.93	181.23	-19.96	ASCENT
26	2026-06-07T00:00:26	0:00:26	655.78	936.97	24.1	8.35	97	8.532037	76.864803	19.26	8.01	184.44	-19.83	ASCENT
27	2026-06-07T00:00:27	0:00:27	674.99	934.71	24.1	8.35	99	8.532075	76.864838	48.68	7.01	187.28	-19.21	ASCENT
28	2026-06-07T00:00:28	0:00:28	692.5	932.83	24	8.34	100	8.532116	76.864865	79.36	8.66	190.32	-17.51	ASCENT
29	2026-06-07T00:00:29	0:00:29	709.36	930.84	23.8	8.34	98	8.532159	76.864896	109.79	7.41	193.24	-16.86	ASCENT
30	2026-06-07T00:00:30	0:00:30	725.52	929.09	23.6	8.34	95	8.532201	76.864925	140.09	7.53	196.24	-16.16	ASCENT
31	2026-06-07T00:00:31	0:00:31	741.4	927.35	23.5	8.34	97	8.532224	76.864958	170.37	7.1	199.57	-15.88	ASCENT
32	2026-06-07T00:00:32	0:00:32	757.04	925.56	23.4	8.34	98	8.532277	76.864985	199.87	8.88	203.04	-15.64	ASCENT

Fig. 5.7 Exported Telemetry CSV File

Packet No	Altitude (m)	Temperature (°C)	Voltage (V)	Mission Phase
1	10.14	28.3	8.40	PRE_LAUNCH
2	10.02	28.4	8.40	PRE_LAUNCH
3	9.81	28.4	8.39	PRE_LAUNCH
4	34.72	28.1	8.39	LAUNCH
5	59.94	28.0	8.39	LAUNCH

Table 5.3 Sample Exported Telemetry Data

The exported telemetry data demonstrates correct mission behavior. During launch and ascent phases, altitude increases steadily while atmospheric pressure decreases. Temperature gradually reduces with increasing altitude. Battery voltage remains stable throughout the mission, indicating healthy payload operation. GPS coordinates continuously update, allowing accurate payload tracking and recovery.

5.15 Overall System Performance

Module	Status
Telemetry Engine	Passed
Simulation Manager	Passed
Zustand Store	Passed
Mission Tracker	Passed
GPS Map	Passed
3D Viewer	Passed
Analytics Dashboard	Passed
Command Panel	Passed
FDD Module	Passed
CSV Export	Passed
Graph Export	Passed

Table 5.4 Overall Performance of the System

All major components functioned successfully during testing.

5.16 Discussion

The results demonstrate that the proposed Ground Control Software successfully integrates telemetry monitoring, mission visualization, GPS recovery, orientation tracking, fault diagnosis, and data analytics within a unified platform.

Compared to traditional educational telemetry systems, the developed solution provides:

- Improved visualization capabilities
- Enhanced mission awareness
- Better recovery support
- Aerospace-inspired operator experience
- Scalable web-based deployment

The modular architecture also enables future integration with actual CanSat flight hardware.

5.17 Chapter Summary

This chapter presented the operational results and performance evaluation of the developed Ground Control Software. The system successfully completed the simulated mission lifecycle while providing real-time telemetry monitoring, GPS tracking, orientation visualization, fault detection, command management, and export capabilities. The next chapter discusses system testing, validation procedures, and performance verification.

CHAPTER 6

TESTING AND VALIDATION

6.1 Introduction

Testing and validation are essential stages in software development to ensure that the developed system performs according to specified requirements. The CanSat Ground Control Software was subjected to comprehensive testing to verify telemetry processing, mission tracking, visualization modules, command execution, data export functionality, and fault monitoring capabilities.

The objective of testing was to evaluate the reliability, accuracy, responsiveness, and operational stability of the system throughout the complete mission lifecycle.

6.2 Testing Objectives

The primary objectives of testing were:

- Verify telemetry reception and processing.
- Validate mission phase transitions.
- Test GPS tracking functionality.
- Evaluate 3D orientation visualization.
- Verify graph generation and updates.
- Validate CSV export functionality.
- Test command execution system.
- Evaluate fault detection mechanisms.
- Assess overall system stability.

6.3 Testing Environment

Hardware Configuration

Component	Specification
Processor	Intel Core i5 or Higher
RAM	8 GB
Storage	10 GB Available
Graphics	Integrated Graphics
Display	Full HD Monitor

Table 6.1 Hardware Configuration

Software Configuration

Component	Technology
Operating System	Windows 11
Frontend	React.js
Build Tool	Vite
State Management	Zustand
Mapping	Leaflet.js
Visualization	Three.js
Analytics	Chart.js
Browser	Google Chrome

Table 6.2 Software Configuration

6.4 Functional Testing

Functional testing was conducted to verify that each module performs its intended operation correctly.

Test Case 1: Telemetry Start and Stop

Objective:
Verify telemetry stream control.

- Procedure:**
1. Start telemetry reception.
 2. Observe packet generation.
 3. Stop telemetry reception.

- Expected Result:**
- Packets generated during active state.
 - Packet generation stops after command execution.

Status: PASSED

Test Case 2: Mission Phase Tracking

Objective:
Validate automatic mission phase transitions.

Expected Sequence:

PRE-LAUNCH
↓
LAUNCH
↓
ASCENT
↓
APOGEE
↓
SEPARATION
↓
DESCENT
↓
RECOVERY
↓
MISSION COMPLETE

Status: PASSED

Test Case 3: Telemetry Dashboard

Objective:

Verify real-time sensor display.

Parameters Tested:

- Altitude
- Pressure
- Temperature
- Voltage
- RSSI
- GPS Coordinates

Result:

All parameters updated successfully.

Status: PASSED

Test Case 4: Chart Visualization

Objective:

Verify dynamic graph updates.

Graphs Tested:

- Altitude vs Velocity
- Temperature vs Pressure
- Voltage vs RSSI

Result:

Continuous graph updates observed.

Status: PASSED

Test Case 5: GPS Tracking System

Objective:

Validate payload tracking.

Verification Points:

- Coordinate updates
- Marker movement
- Trajectory generation
- Recovery path visualization

Status: PASSED

Test Case 6: 3D Orientation Viewer

Objective:

Validate orientation rendering.

Parameters Tested:

- Roll
- Pitch
- Yaw

Result:

Smooth animation and real-time updates achieved.

Status: PASSED

Test Case 7: Command Uplink System

Objective:

Verify mission command execution.

Commands Tested:

- Manual Separation
- Emergency Parachute
- GPS Recalibration
- Beacon Activation
- Redundant Bus Activation

Result:

Commands executed successfully after safety validation.

Status: PASSED

Test Case 8: Fault Detection System

Objective:

Verify DGSR monitoring.

Faults Tested:

- Excessive Descent Rate
- GPS Failure
- Separation Failure
- Recovery Beacon Activation

Result:

Fault indicators updated correctly.

Status: PASSED

Test Case 9: CSV Export

Objective:

Validate telemetry export.

Result:

Complete mission telemetry successfully exported.

Status: PASSED

Test Case 10: Graph Export

Objective:

Verify graph image generation.

Result:

Telemetry charts exported successfully.

Status: PASSED

6.5 System Test Matrix

Test ID	Module	Expected Result	Status
TC-01	Telemetry Engine	Real-time packets	Passed
TC-02	Mission Tracker	Correct phase transitions	Passed
TC-03	Dashboard	Real-time updates	Passed
TC-04	Analytics	Dynamic graphs	Passed
TC-05	GPS Map	Accurate tracking	Passed
TC-06	3D Viewer	Correct orientation	Passed
TC-07	Commands	Successful execution	Passed
TC-08	Fault Monitor	Correct fault detection	Passed
TC-09	CSV Export	Complete data export	Passed
TC-10	Graph Export	Image generation	Passed

Table 6.3 System Test Matrix

6.6 Performance Evaluation

Telemetry Performance

Parameter	Observation
Update Rate	1 Hz
Packet Loss	None
Response Time	Instantaneous
Data Consistency	High

Table 6.4 Performance Evaluation

User Interface Performance

Parameter	Result
Responsiveness	Excellent
Rendering Speed	Smooth
Navigation	Easy
Visualization Quality	High

Table 6.5 UI Performance

6.7 Reliability Analysis

The system maintained stable operation throughout the mission.

Reliability Indicators

- No telemetry interruption.
- No application crashes.
- Consistent GPS tracking.
- Accurate graph rendering.
- Stable state management.

The centralized Zustand architecture significantly improved reliability and synchronization.

6.8 Validation of Mission Objectives

Objective	Validation Status
Real-Time Telemetry	Achieved
Mission Tracking	Achieved
GPS Recovery	Achieved
3D Visualization	Achieved
Data Analytics	Achieved
Fault Monitoring	Achieved
Command Management	Achieved
Data Export	Achieved

Table 6.6 Validation Status

All project objectives were successfully achieved.

6.9 Advantages Observed During Testing

The developed system demonstrated several advantages:

1. Fast telemetry processing.
2. Modern aerospace-style dashboard.
3. Effective recovery tracking.
4. Comprehensive mission awareness.
5. Modular architecture.
6. Scalable implementation.
7. Easy deployment.

8. User-friendly interface.

6.10 Limitations

The current implementation has the following limitations:

1. Uses simulated telemetry instead of live flight hardware.
2. Limited multi-user support.
3. No cloud-based telemetry storage.
4. No AI-assisted anomaly prediction.
5. Limited communication protocol integration.

These limitations provide opportunities for future enhancement.

6.11 Chapter Summary

This chapter presented the testing and validation results of the CanSat Ground Control Software. All functional modules including telemetry monitoring, mission tracking, GPS recovery, 3D visualization, fault diagnosis, command execution, and export services successfully passed testing. The results confirm that the system meets its design objectives and is suitable for educational CanSat missions, aerospace demonstrations, and telemetry monitoring applications.

CHAPTER 7

FUTURE SCOPE

7.1 Introduction

The developed CanSat Ground Control Software successfully demonstrates real-time telemetry monitoring, mission tracking, GPS recovery support, fault diagnosis, and aerospace visualization. However, advancements in aerospace systems, satellite communication, artificial intelligence, and cloud technologies provide numerous opportunities for future enhancement.

This chapter discusses potential improvements and future developments that can increase the capabilities and operational effectiveness of the proposed system.

7.2 Integration with Real Hardware

The current implementation utilizes a simulation engine for telemetry generation.

Future versions can be integrated with:

- ESP32 Microcontrollers
- Arduino-Based Flight Computers
- STM32 Controllers
- Raspberry Pi Payload Systems
- LoRa Communication Modules
- XBee Radio Systems

Benefits:

- Real-world deployment
- Hardware validation
- Flight-ready operation

7.3 Artificial Intelligence Integration

Artificial Intelligence can significantly enhance mission operations.

Potential Applications

Predictive Fault Detection

AI models can predict:

- Sensor failures
- Battery degradation
- Communication loss

- System anomalies

Mission Risk Assessment

Machine learning algorithms can analyze telemetry patterns and identify potential mission risks before failures occur.

Benefits:

- Improved reliability
- Reduced mission risk
- Automated monitoring

7.4 Cloud-Based Telemetry Storage

Currently, telemetry data is stored locally.

Future enhancements may include:

- AWS Cloud Integration
- Google Cloud Storage
- Microsoft Azure Services
- Distributed Telemetry Databases

Advantages:

- Remote monitoring
- Long-term data retention
- Multi-user access
- Mission analytics

7.5 Multi-User Ground Station

The current system is designed for a single operator.

Future versions can support:

- Multiple operators
- Role-based access
- Collaborative mission monitoring
- Shared telemetry streams

Benefits:

- Better mission coordination
- Improved operational efficiency

7.6 Advanced Communication Systems

Future implementations may support:

LoRa Communication

Benefits:

- Long-range telemetry
- Low power consumption

Satellite Communication

Benefits:

- Global coverage
- Beyond-line-of-sight operation

Mesh Networking

Benefits:

- Improved redundancy
- Fault tolerance

7.7 Mobile Application Development

A dedicated mobile application can be developed for:

- Telemetry monitoring
- Alert notifications
- Payload recovery
- Mission status updates

Advantages:

- Portability
- Accessibility
- Faster response

7.8 Enhanced 3D Visualization

Future versions may include:

- Digital Twin Technology
- Complete 3D Mission Environment
- Terrain Visualization
- Augmented Reality Interfaces

Benefits:

- Better operator awareness
- Advanced mission simulation

7.9 Autonomous Recovery System

Future recovery systems may incorporate:

- AI-Based Recovery Prediction
- Automated Landing Zone Detection
- Autonomous Search Algorithms
- Drone-Assisted Recovery

Benefits:

- Faster payload retrieval
- Reduced operational effort

7.10 Advanced Analytics Dashboard

Future analytics capabilities may include:

- Real-Time Predictive Models
- Flight Performance Analysis
- Sensor Health Monitoring
- Automated Report Generation

Benefits:

- Enhanced decision support
- Better mission evaluation

7.11 Smart Mission Control Platform

The long-term vision is to transform the Ground Control Software into a complete aerospace mission management platform capable of supporting:

- CanSat Missions
- CubeSat Missions
- High-Altitude Balloons
- UAV Operations
- Scientific Payload Monitoring

7.12 Chapter Summary

Future developments such as AI integration, cloud-based telemetry storage, advanced communication systems, digital twin technology, and autonomous recovery mechanisms can significantly enhance the capabilities of the proposed Ground Control Software. These

advancements will enable the system to evolve into a comprehensive aerospace mission management platform.

CHAPTER 8

CONCLUSION

The objective of this project was to design and develop an ISRO-inspired CanSat Ground Control Software capable of providing real-time telemetry monitoring, mission visualization, payload tracking, fault diagnosis, and recovery support.

The developed system successfully integrates multiple aerospace monitoring components into a unified web-based platform. Using React.js, Vite, Zustand, Three.js, Leaflet.js, and Chart.js, a modern mission operations dashboard was created that provides operators with comprehensive situational awareness throughout the complete mission lifecycle.

The system supports real-time visualization of critical mission parameters including altitude, pressure, temperature, battery voltage, RSSI, GPS coordinates, and inertial sensor orientation. The implementation of GPS-based recovery tracking, telemetry analytics, mission phase monitoring, and fault detection significantly improves mission management capabilities.

The simulation engine successfully modeled all eight mission phases, enabling realistic mission execution and system validation. Testing results confirmed that all major modules including telemetry monitoring, mission tracking, command management, analytics, export services, and fault diagnosis operated reliably and achieved their intended objectives.

The modular software architecture ensures scalability and enables future integration with actual CanSat hardware platforms. Furthermore, the web-based design provides accessibility, ease of deployment, and cross-platform compatibility.

In conclusion, the developed Ground Control Software successfully demonstrates how modern web technologies can be applied to aerospace mission operations, educational satellite programs, telemetry visualization systems, and payload recovery applications. The project provides a strong foundation for future enhancements involving artificial intelligence, cloud computing, advanced communication systems, and real-world flight hardware integration.

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